

A COMPLETE BOOK OF TRAIL RUNNING PREPARATION

Flat to Mountain

The reference for trail runners who train on flat terrain. The complete flat-terrain workout library, the methodology behind it, three full training plans, mountain legs built indoors, food worth cooking, and a rehearsed race day.

FLAT TRAINING



MOUNTAIN RACING

THIRD EDITION · EXPANDED & REVISED

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PART I

The Method

Why mountains hurt, what elevation actually trains, and how to substitute it honestly. Chapters 1–3: the flatlander's problem, the physiology behind every session in this book, and the zones and benchmark tests that anchor them.

How to Use This Book

This book has one job: to get a runner who trains on flat terrain to the finish line of a mountain trail race — strong, fed, and smiling. It is organised so you can read it cover to cover once, then live inside three or four chapters for fourteen weeks.

The spine of the book is a simple argument: flat terrain is not a handicap, it is a different syllabus. Part I teaches the syllabus — why mountains hurt, what elevation actually trains, and how to measure yourself honestly. Part II is the workout library: every flat-terrain session worth knowing, organised by what it builds. Part III builds mountain legs indoors and teaches the technique that multiplies fitness. Part IV gives you complete training plans and — new in this edition — the rules for building your own. Parts V and VI cover the support systems (food, environment, injury, recovery) and the race itself. The appendices are for the fridge door.

If you are starting a plan this week, read Chapters 1–3, pick a plan in Part IV, and read the fueling chapter (Chapter 17) before your first long run. Everything else can be absorbed as the weeks pass.

Choosing your plan

QUESTION	PLAN A (CH. 14)	PLAN B (CH. 15)	PLAN C (CH. 16)
Runs per week	5–6	4	5–6
Peak weekly volume	~58–62km	~44–48km	~55–60km
Weekly time (incl. strength)	7–9 hours	5–6 hours	7–8 hours
Length	14 weeks	14 weeks	10 weeks
Best for	Experienced runners, ambitious goals	Busy lives, injury-prone, first mountain race	Already-fit runners short on calendar
Finish outcome	Racing the course	Finishing strong and enjoying it	Racing on borrowed time

All plans share the same skeleton — two quality days, one long run, scheduled recovery weeks — and all will get you to the finish. Be honest about your available hours and your injury history, pick one, and **do not hop between them**. A modest plan completed beats an ambitious plan abandoned, every single time.

The five rules that govern everything

1. **Consistency beats heroics.** Fourteen unremarkable weeks build a remarkable race.
2. **Easy means easy.** If your easy runs are quietly medium, your hard sessions will be quietly mediocre.
3. **Strength work is a session, not a garnish.** Skipping it is skipping elevation training.
4. **Every long run from week 6 is a rehearsal** — of food, gear, pacing, and patience.
5. **Arrive at the start line slightly undertrained rather than slightly injured.** Nobody has ever regretted the first; everyone has regretted the second.

Effort zones (Z1–Z6) are defined in Chapter 3 and used everywhere. When the book says "Z2", it means a specific, honest, conversational effort — anchor the zones to a recent 5k or 10k before week 1.

Chapter 1 — The Flatlander's Problem

Training for a mountain trail race while living somewhere pancake-flat feels like preparing for a swim meet in a desert. Your race will throw ~1,000m of climbing and descending at you across 28–30km of technical terrain. Your daily training ground offers canal paths, park loops, and the occasional bridge.

Here is the good news: **flat-terrain runners routinely finish — and race well — in the mountains.** But only if they train deliberately. The flatlander who simply logs easy kilometres arrives at the start line with a big aerobic engine bolted to legs that have never experienced sustained climbing load or eccentric descending damage. The flatlander who trains smart arrives with three weapons:

1. **Superior leg speed and running economy** — flat terrain is the best place in the world to get fast.
2. **A muscular system prepared for climbing** through strength work and elevation substitutes.
3. **A fueling and pacing strategy rehearsed to perfection**, because flat long runs make it easy to practice eating and pacing without terrain chaos.

This book is built around that philosophy. You cannot replicate a mountain in a flat city — so don't try to fake it badly. Instead, **overdevelop what flat terrain gives you (speed, economy, endurance) and specifically target what it doesn't (climbing strength, descending resilience, hiking economy).**

What elevation actually trains — and how we substitute it

MOUNTAIN DEMAND	WHAT IT DOES TO YOU	FLAT-TERRAIN SUBSTITUTE
Sustained climbing	High muscular force at low cadence; big glute/calf load	Stair repeats, treadmill incline, weighted step-ups, bike/stair machine
Steep descending	Eccentric quad damage, braking control	Downhill treadmill, fast strides, heavy slow squats, plyometrics
Power hiking	Strength-endurance in glutes and hips	Weighted-vest walking, lunges, step-up volume
Variable pace/effort	Constant gear-shifting	Fartlek, progressive runs, surges within long runs
Technical footing	Ankle stability, proprioception	Trail-ish surfaces (forest paths, sand, grass), single-leg balance work
Long time on feet	Durability at 3–5 hours	Long runs by <i>time</i> , back-to-back weekend runs

The single biggest mistake flat-terrain trail runners make is training only for the fitness component and ignoring the **musculoskeletal** component. A 4-hour mountain race destroys quads, not lungs. Every week of this book's plans includes strength work for exactly this reason. Treat it as a workout, not an optional extra.

Race effort for a 28–30km mountain trail is mostly Zone 2 with Zone 3 moments. That surprises people. The speed work in this book is not for racing fast — it's for making Zone 2 *faster and cheaper*, so that your easy effort on race day covers more ground per heartbeat.

The flatlander's honest audit

Before week 1, answer these on paper. They decide which chapters matter most to you:

- **What is the tallest continuous climb near me?** A 12-storey stairwell, a stadium, a motorway bridge, a treadmill? That's your mountain. Chapter 8 turns it into a training venue.
 - **What is my longest recent run?** If under 90 minutes, spend two to four weeks building the long run before starting a plan — the plans assume you can run 12–14km continuously.
 - **Have I ever done eccentric strength work?** If no, start Strength B (Chapter 10) at half doses. The soreness of the first two weeks is the adaptation arriving; don't schedule it before a long run.
 - **What does the race roadbook say?** Vert, longest climb, longest descent, aid stations, cutoffs, mandatory kit. Read it now, not in race week. Every number in it should shape a training decision.
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Chapter 2 — The Physiology of Substitution

You don't need a physiology degree to train well, but you need to know *why* each session exists — because on a tired Tuesday, "why" is what keeps the session honest. This chapter is the why, in plain language.

The four systems that decide your race

1. The aerobic engine (mitochondria, capillaries, stroke volume). Built by volume at easy effort, over months. It is the single biggest determinant of a 3–5 hour race and the least glamorous to train. Every easy kilometre is a deposit. This is why 80% of your training time stays in Z1–Z2, and why "easy means easy" is rule number two: the adaptations of easy running happen *at easy effort*, and running medium instead simply borrows recovery from your hard days without adding anything.

2. The lactate threshold — your sustainable ceiling. The effort you can hold for roughly an hour. Raise it, and every effort below it — including all-day race effort — gets cheaper. Threshold work (Chapter 6) is the highest-value intensity for a mountain 30k because long climbs are, functionally, threshold intervals with scenery.

3. VO2max and running economy — the roof and the fuel bill. VO2max sets the roof over everything; economy is how much of it you spend per kilometre. Flat terrain is the world's best laboratory for both. Short fast repeats and strides (Chapters 4–5) teach your legs to recycle elastic energy, which is free speed on every runnable section of the course. **Economy is the flatlander's secret weapon:** mountain locals get strong legs from terrain; almost none of them do structured speed work. You will.

4. Musculoskeletal durability — the one that actually ends races. Tendons, quads, calves, and connective tissue adapt slower than the cardiovascular system — weeks to months, not days. Climbing loads calves and glutes concentrically; descending loads quads *eccentrically*, which is the muscle working while lengthening, like lowering a heavy box slowly. Eccentric load is what causes the trashed-quads, walking-backwards-downstairs feeling after a mountain race. It is also remarkably trainable.

The repeated bout effect — your descent insurance

Here is the most useful piece of sports science in this book: **a single exposure to eccentric load protects you from the same load for weeks.** It's called the repeated bout effect. Do one bout of unaccustomed downhill running or heavy eccentric strength work and you'll be sore for two days — but the *next* time, the same dose produces dramatically less damage, and the protection lasts four to six weeks.

For the flatlander this changes everything. You cannot train descents weekly, but you don't need to. You need **regular eccentric strength work (Strength B, Chapter 10)** plus **a few concentrated eccentric exposures** — a downhill treadmill session, a session of drop jumps and step-downs, or one real-trail day — spaced through the plan, with the last one landing two to three weeks before the race. That schedule is already built into the plans. Trust it, and don't "bonus" extra eccentric work in race month.

Hiking is an energy system too

Above roughly 12–15% gradient, walking is faster per unit of energy than running — for everyone, including the elites you'll watch power-hike away from you. Hiking economy is trainable and almost nobody trains it. The weighted hikes and stair sessions in this book are not filler; they are the difference between climbing at a sustainable cost and arriving at the summit with your race already spent.

What this means for your week

SESSION TYPE	SYSTEM TRAINED	FREQUENCY IN THE PLANS
Easy runs (Z1–Z2)	Aerobic engine, durability	3–4 per week, ~80% of time
Threshold work (Z4)	Sustainable ceiling	~1 per week
VO2max / speed (Z5–Z6)	Roof + economy	~1 per week
Strides & drills	Economy, form	2 per week, nearly free
Stairs / incline / hikes	Climbing strength, hiking economy	1–2 per week
Strength A & B	Climbing force, descent armour	2 per week
Long run	Everything at once	1 per week, the cornerstone

Every session in Part II names the system it trains. When you must cut something from a busy week, cut with this table in view: the long run and strength work go last.

Chapter 3 — Zones, Benchmarks & Testing

All sessions in this book are described by effort. Pace is a *result* of effort applied to terrain; on a mountain it is meaningless as a target. Learn the zones by feel on flat ground, where they're easy to calibrate, and they'll transfer to any gradient.

The effort zones

ZONE	NAME	FEEL	ROUGHLY
Z1	Recovery	Conversational, almost embarrassingly slow	5k pace + 2:00–2:30/km
Z2	Easy	Full sentences, nose-breathing possible	5k pace + 1:15–2:00/km
Z3	Steady / Marathon effort	Short sentences	5k pace + 0:45–1:10/km

ZONE	NAME	FEEL	ROUGHLY
Z4	Tempo / Threshold	A few words at a time, "comfortably hard"	5k pace + 0:15–0:35/km
Z5	Interval / VO2max	One-word answers	~5k pace
Z6	Repetition / Sprint	No talking, near-max	Faster than 5k pace

The talk test outranks the watch. If your watch says Z2 but you can't speak in full sentences, believe your mouth.

Anchoring the zones: the 5k time trial

Before week 1, run a 5k time trial — a parkrun is perfect, a measured solo effort works. Warm up 15 minutes with strides, then race it honestly. Your finish time anchors every pace in the table above and gives you the baseline for retests.

If a 5k feels too daunting to start, a **30-minute solo test** works: run 30 minutes at the hardest effort you can hold evenly; your average pace for the final 20 minutes approximates threshold (Z4), and the zones can be hung off that instead.

The flatlander's benchmark battery

Retest in the plan's recovery weeks (weeks 4 and 8) — recovery weeks have room for a test, and the numbers tell you whether the plan is working before the race does.

BENCHMARK	PROTOCOL	WHAT IT TELLS YOU
5k time trial	All-out, rested, measured course	Anchors all zones; tracks speed & threshold
Stair benchmark	Fixed staircase or stair machine, 10 min steady Z3: count floors/metres	Climbing fitness, hiking economy
HR drift check	60 min steady Z2, compare HR of first vs. second 30 min	Aerobic base: drift under ~5% = solid
Long-run recovery	How do legs feel 36–48h after the long run?	Durability; readiness to add load
In-race carb tolerance	g/hour comfortably absorbed on long runs	Fueling ceiling (Chapter 17 raises it)

Improvement will be lumpy. Expect the stair benchmark and carb tolerance to improve fastest — they start from the least training. Expect the 5k to improve modestly; this is not a 5k plan. The number that matters most is the least measurable: how ordinary a 20km long run starts to feel.

Heart rate, GPS, and honest instruments

- **If you train by HR:** anchor zones off the time trial (or a lab/field LTHR test), not off 220-minus-age, which is a coin toss. Cardiac drift, heat, and caffeine all move HR; treat it as a second opinion, not a verdict.
- **GPS pace lies** on tight loops, under trees, and always on stairs. Effort and time are the currencies of this book; pace is for flat track sessions only.
- **The one log that matters:** after each key session, one line — what you did, how it felt. In week 14, that ledger out-argues every taper doubt (Chapter 21).

PART II

The Workout Library

The complete catalogue of flat-terrain sessions, organised by the system each one trains. Chapters 4–9: neuromuscular speed, VO2max, threshold, the long run library, the elevation substitutes, and the fartlek & compound sessions.

Speed work is where flat terrain becomes an advantage. This part is a complete library of flat-terrain sessions, organised by the system each one trains. The plans in Part IV draw from it by name; when you outgrow the plans, you'll build your own weeks from these pages.

How to read a session entry. Each session lists its structure, why it works for trail, and how to progress it. Progress **one dial at a time** — more reps, *or* shorter rest, *or* faster pace, never all three in the same week.

The universal warm-up before anything at Z4 or faster: 10–15 minutes easy jog, then drills (high knees, butt kicks, skips, leg swings), then 3–4 strides building to the session's pace. Cold sprinting is how flatlanders donate hamstrings to the cause.

The weekly speed-work rules:

1. **Two quality days per week, never back-to-back.** Typically Tuesday (speed) and Thursday (tempo or stairs).
2. **80/20 still applies.** Roughly 80% of weekly time stays Z1–Z2. Speed work is potent precisely because it's scarce.
3. **When in doubt: cut reps, not intensity, for sprints — and cut intensity, not reps, for long runs.**
4. **A session you can't finish with good form is a session that should have ended two reps ago.**

Chapter 4 — Neuromuscular Speed: Strides, Sprints & Elastic Legs

System trained: economy, elastic recoil, top-end speed. Cheap to add, criminally underused by trail runners.

Short, fast running teaches tendons to store and return energy — which lowers the cost of **every** stride you take for the rest of your life, including the slow ones up a mountain. These sessions are short on volume and high on quality. Full recoveries. If you're panting into the next rep, you're doing a different (worse) session.

4.1 Strides — the daily vitamin

- **Structure:** 4–8 × 20–30s relaxed accelerations to ~Z6 at the end of easy runs; walk/jog back full recovery.
- **Why it works for trail:** nearly free speed and coordination. Twice a week, forever.
- **Form cue:** tall, quick, quiet. Think "fast", never "hard" — a stride is speed without strain.
- **Variants:** *grass strides barefoot* (foot strength, proprioception); *downhill strides* on any gentle slope at play-speed (descent rehearsal, see 8.5); *uphill strides* on a bridge ramp (power).

4.2 Flying 30s

- **Structure:** 6–8 × 30m at near-max, entered from a 30m rolling build; walk back 2–3 min. Total sprint volume tiny, quality total.
- **Why it works for trail:** pure mechanics and recruitment. The nervous system learns to fire everything; every submaximal pace gets cheaper.
- **When:** in place of strides once every week or two, on fresh legs only.

4.3 Track Attack — 400m repeats

- **Structure:** 10–16 × 400m at Z5–Z6, with 200m jog or 60–90s standing recovery.

- **Why it works for trail:** short, fast repeats build raw leg speed, elastic recoil, and neuromuscular sharpness. On the trail this translates into cheaper strides on runnable sections and the ability to change gear instantly when terrain changes.
- **Progression:** early — 10 × 400m @ Z5, 90s rest; mid — 12 × 400m @ Z5–Z6, 75s rest; peak — 16 × 400m alternating Z5 and Z6, 60s rest.
- **Form cue:** tall posture, quick feet, land under your hips. Sloppy 400s are wasted 400s.

4.4 200m turnover session

- **Structure:** 12–16 × 200m at Z6 (mile-race feel, not sprint), 200m jog between; optionally in 2 sets with 3 min between.
- **Why it works for trail:** cadence and turnover without the fatigue cost of 400s. Excellent in taper weeks (short version: 6–8 reps) to stay sharp while shedding fatigue.

4.5 Stair sprints — the flatlander's hill sprint

- **Structure:** 8–12 × 20–30s maximal stair climbs (or steepest ramp available), walk down + stand until fully recovered (~2 min). See Chapter 8 for venue-finding.
- **Why it works for trail:** maximal-force climbing recruitment — the closest thing to hill sprints without a hill. Doubles as ankle-stiffness and calf-power training.
- **Caution:** the most potent Achilles-loading session in the book. Introduce with 4–6 reps and never in the same week as a big jump in stair volume.

The taper truth about speed: neuromuscular sharpness fades in days, fitness fades in weeks. This is why strides and short reps survive into race week while everything else shrinks.

Chapter 5 — VO2max: Raising the Roof

System trained: maximal aerobic power — the roof under which threshold and race effort live. Hard, effective, and to be respected.

VO2max sessions accumulate 12–20 minutes of work at ~Z5 (about 5k effort). They hurt in a productive, controlled way. One per week in build phases is plenty; the plans place them on Tuesdays.

5.1 The Kilometre Club — 1km repeats

The bread-and-butter session of this whole programme.

- **Structure:** 5–8 × 1km at Z4–Z5, with 90 seconds to 2:30 jog recovery.
- **Why it works for trail:** 1km is long enough to train sustained hard effort (like a 6–8 minute climb) but short enough to hold quality form. The repeated surging and settling mimics the rhythm of a hilly course far better than one continuous tempo.
- **Progression:** early — 5 × 1km @ Z4, 2:30 jog; mid — 6–7 × 1km @ Z4–Z5, 2:00 jog; peak — 8 × 1km @ Z5, 90s jog.
- **Trail twist:** run the last 200m of every repeat with exaggerated knee drive and arm swing — you're rehearsing the muscular pattern of attacking a climb.

5.2 800s (the classic)

- **Structure:** 6–10 × 800m @ Z5, jog 400m (or equal time) between.
- **Why it works for trail:** the sweet spot between quality and volume; each rep is a 3–4 minute climb in disguise. If Kilometre Club reps keep collapsing in the final 200m, drop to 800s for a few weeks and return.

5.3 Billat 30/30s — VO2max for people who hate track pain

- **Structure:** after warm-up, alternate 30s @ Z5 / 30s easy jog, continuously, for 12–20 minutes. (60/60s are the bigger sibling.)
- **Why it works for trail:** the short floats keep oxygen uptake pinned near max while the legs get micro-breaks — big stimulus, surprisingly civilised suffering, and no track needed; any park path works.
- **Bonus:** the constant gear-changing is the most race-like rhythm in this chapter.

5.4 The Ladder

- **Structure:** 400–800–1200–800–400 @ Z5, jog half the rep distance between. (Pyramid to 1600 for the ambitious.)
- **Why it works for trail:** climbing a real mountain is never metronomic — efforts lengthen and shorten. Ladders train you to commit to a hard effort *without knowing exactly when it ends*, which is nine-tenths of mountain psychology.

5.5 3-minute repeats (effort-based, anywhere)

- **Structure:** 5–7 × 3:00 @ Z5 by feel, 2:00 jog. No track, no GPS accuracy needed.
- **Why it works for trail:** the purest effort-based VO2 session — perfect for wind, bad weather, or unmeasured paths, and it teaches the internal pacing clock you'll race by.

5.6 Cut-downs

- **Structure:** 6 × 1km starting @ Z4 and finishing @ Z5+, each rep slightly faster, 2:00 jog.
- **Why it works for trail:** finishing faster than you start is the signature of every well-raced mountain course. Practise the *feeling* of acceleration on tired legs in week 9 and it will be there at kilometre 26.

Choosing within this chapter: Kilometre Club and Track Attack are the plan staples. Swap in 30/30s when you have no measured course, ladders when you're bored, 800s when form is fraying, cut-downs late in the build. Never two of these in the same week.

Chapter 6 — Threshold: Raising the Floor

System trained: the lactate threshold — your sustainable ceiling, and the single highest-value intensity for a 3–5 hour race.

Threshold work lives at Z4: "comfortably hard", a few words at a time, roughly your best one-hour effort. Total Z4 time per session: 20–40 minutes. It should end feeling like you had 10 more minutes in you. If a tempo turns into a time trial, it has stopped training the threshold and started spending it.

6.1 The classic tempo

- **Structure:** 20–40 min continuous @ Z4, inside an easy-run sandwich.
- **Why it works for trail:** the long-climb simulator. A 35-minute tempo is your race's biggest climb, minus the view.
- **Progression:** 20 → 25 → 30 → 35 → 40 minutes across a block, then reset shorter and slightly faster.

6.2 Cruise intervals

- **Structure:** 3–5 × 8–12 min @ Z4, 2 min jog between.
- **Why it works for trail:** more total quality than a continuous tempo with better form retention. The 2-minute floats mimic the false flats between climbs. The plans' "3×12 @ Z3" and "4×10 @ Z3–Z4" long-run blocks are this session folded into a long run.

6.3 Wave tempo (over-unders)

- **Structure:** 4–6 × (3 min slightly over threshold, Z4+ / 3 min slightly under, Z3–Z4), continuous.
- **Why it works for trail:** mountains never let you settle. Over-unders train the exact metabolic skill a rolling course demands: absorbing a surge *without stopping to recover from it*. The most race-specific threshold session in the book.

6.4 Progressive tempo

- **Structure:** 30–40 min starting @ Z3 and finishing @ Z4+, a smooth ramp, no watch-staring — let effort drift up every 10 minutes.
- **Why it works for trail:** teaches negative-split discipline at medium cost. Also the best "I feel flat today" quality session — the gentle start recruits you into it.

6.5 Broken tempo for the time-crunched

- **Structure:** 2 × 15 min @ Z4, 3 min jog between — or 10 min @ Z4 at the start *and* end of a 50-minute run.
- **Why it works for trail:** fits real life. The second block on slightly tired legs quietly rehearses race-day fatigue resistance.

6.6 The hour of power (peak-phase capstone)

- **Structure:** 3 × 15 min @ Z4, 3 min jog — done once, in week 9–11, as a fitness graduation exam.
- **Why it works for trail:** 45 minutes of quality is the biggest sustainable-effort dose in the book. Pass this and no single climb on the course can intimidate you. Requires a recovery day either side; do not repeat weekly.

Chapter 7 — The Long Run Library

System trained: everything at once. The cornerstone. If a week collapses, save the long run.

Long runs are listed in the plans by distance, but from week 6 the distance is the least important thing about them: **every long run from week 6 onward is a rehearsal** — of fueling (Chapter 17), gear, pacing, and patience. Run them by time when the watch taunts you: a mountain 30k is a 3.5–5 hour event; a 24km flat long run at ~2.5 hours is the honest equivalent.

7.1 The Negative Split — the most race-specific session a flatlander can do

- **Structure:** a long run divided in thirds — first third Z1–Z2, middle third Z2, final third Z3 (finishing the last 2–3km near Z4 in peak weeks).
- **Why it works for trail:** a 28–30km mountain race is won or lost in the final hour, on trashed legs. Progressive long runs teach you to run *strong on tired legs* — the exact skill descending and running the final flat kilometres demands.
- **Golden rule:** if you can't speed up in the final third, you started too fast. This is also the number-one race-day lesson.

7.2 The Sandwich long run

- **Structure:** long run with quality blocks folded in — e.g. 20km with 3 × 15 min @ Z3, or 22km with 4 × 10 min @ Z3–Z4, floats easy between.
- **Why it works for trail:** climbs arrive mid-race, not fresh. Threshold blocks at kilometre 15 of a long run are climbing rehearsal with the fatigue included. The plans schedule these in weeks 7 and 11.

7.3 The Surge long run

- **Structure:** easy long run with 8–12 × 1 min @ Z4 sprinkled at random (every time you pass a bridge, a red kite, whatever) — return to Z2 between.
- **Why it works for trail:** the mountain decides when you surge, not you. Random surging teaches the metabolic gear-change without the structure your brain would otherwise negotiate with.

7.4 Back-to-back weekend

- **Structure:** the plan's Saturday long run followed by a Sunday medium run (e.g. 20km + 10km), both easy.
- **Why it works for trail:** starting a run on pre-fatigued legs is the cheapest way to buy "hours 4 and 5" durability without a 5-hour run's injury bill. The plans use these sparingly (weeks 7 and 9 of Plan A) — they cost real recovery.

7.5 The Fast-Finish

- **Structure:** easy long run with the final 3–5km @ Z3–Z4, committed.
- **Why it works for trail:** a smaller, sharper Negative Split. Also the best rehearsal of the mental gearshift at "everything hurts, 5k to go".

7.6 Time-on-feet & the train-ride long run

- **Structure:** occasionally, replace pace entirely: 3+ hours by time on the softest, roughest ground available; hike anything that resembles a slope; practise eating.

- **Why it works for trail:** feet, ankles, and morale adapt to hours, not kilometres. And **once or twice in the plan, travel** — one train ride to real trails, dunes, or a big park with actual gradient teaches feet more than five identical park loops.

7.7 Run/walk rehearsal (Galloway racers)

If you race with a run/walk strategy, rehearse your intervals in every long run from week 6 onward — e.g. 4:00 run / 0:45 walk on flats, switching to effort-based hiking on anything steep. The plans are fully compatible; apply your intervals to the easy and long-run portions. **Walk breaks are a strategy, not a failure** — used from the start, they're worth many minutes at the end.

What long runs are not: weekly death marches. If a long run needs three rest days, it was a race. The Sunday question — "could I do most of it again?" — should usually be answered *yes, grudgingly*.

Chapter 8 — The Elevation Substitutes

System trained: climbing strength, hiking economy, descent resilience. This chapter is the mountain you build out of urban furniture.

8.1 Finding your mountain

Rank of substitutes, best first:

1. **Treadmill at 10–15% incline** — precise, repeatable, weatherproof. The gold standard substitute.
2. **Long staircases** — apartment blocks, stadiums, parking garages, bridges with stair flights. Free, brutally effective, and they train descent (the walk down) too.
3. **Stair machine** — continuous climbing without the descent; ideal for endurance doses.
4. **Bridges, overpasses, dune paths, parking-garage ramps** — short, but shameless repetition fixes short.
5. **Bike or elliptical, big gear, low cadence** — decent leg strength-endurance, zero impact; the cross-training fallback (see 8.6).

Combine two. No single substitute covers everything.

8.2 Stairway A — Power

- **Structure:** 10–15 × 30–60s hard stair climbs, walk down for recovery.
- **Why it works for trail:** maximal climbing force and turnover. The descent between reps is stealth eccentric training — take it seriously, don't clatter down.

8.3 Stairway B — Endurance

- **Structure:** 20–40 minutes of continuous stair climbing / stair machine at Z3, taking steps singly and doubly in alternating blocks.
- **Why it works for trail:** sustained climbing is the race's core demand; this is its purest substitute. Alternating single/double steps varies cadence and force exactly like a real climb's changing gradient.

8.4 Stairway C — The weighted hike

- **Structure:** stairs or steep treadmill (10–15% incline) with a loaded backpack (4–6kg), 30–45 minutes of brisk power hiking.
- **Why it works for trail:** this is *the* session for race-day hiking economy — you **will** hike the steep climbs, and hiking is a trainable skill. Practise the two hiking gears from Chapter 11 here, plus run-hike transitions: 2 minutes climbing hard, 1 minute brisk hiking, repeat, without drama.
- **Poles:** if your race allows them and you'll use them, this is where they get trained (Chapter 11).

8.5 The descent problem — training down without a down

Descending is the flatlander's biggest gap, because even stairs give only short doses. The full arsenal:

- **Downhill treadmill:** if your gym's treadmill declines (–2 to –5%), run easy 10–20 min doses at Z2, fortnightly from week 5, last dose ~2.5 weeks out. Small doses, big protection (repeated bout effect, Chapter 2).
- **Eccentric strength (Strength B, Chapter 10):** slow 5-second squat lowerings, box step-DOWNS, drop jumps — the gym analogue of descent damage, on tap year-round.
- **Downhill strides:** on any gentle slope, grass bank, or dune — fast, playful, quick feet, 6–8 × 20s. Rehearses the braking pattern and the confidence.
- **Stair descents:** every walk down in Stairway A counts. Late in the plan, jog some descents (carefully) — quick, quiet feet.
- **One real-trail day** (Chapter 7.6): the single best descent lesson available. Schedule it weeks 8–11.

8.6 Cross-training as elevation

Ranked substitutes when running load is maxed or a niggle needs a detour: hard cycling in a big gear at low cadence (~60rpm) for climbing strength-endurance; stair machine (already covered); uphill treadmill walking. Thirty to sixty minutes at Z2–Z3 substitutes an easy run almost 1:1 for climbing purposes — this is also the injury-detour menu from Chapter 19.

8.7 Sand, dunes & rough ground

- **Structure:** one run every week or two on the most technical flat ground you have — dry sand, dunes, forest floor, rooty park paths. Easy pace; the terrain provides the intensity.
- **Why it works for trail:** builds ankle strength, forces quick light steps, and punishes overstriding. Dune running is honorary hill training. Barefoot grass strides after, for dessert.

8.8 Durability extras that cost almost nothing

- **Walk more.** Weighted-vest walks, long city walks, taking every staircase you meet. Time on feet is a stimulus.
- **Run on the softest, roughest ground available** for easy runs. Ankles and stabilisers adapt to chaos only by experiencing chaos.
- **Practice power-hiking form now:** hands on knees for very steep pitches, shorter quicker steps, forward lean from the ankles (full technique in Chapter 11).

Chapter 9 — Fartlek & Compound Sessions

System trained: gear-changing — the mountain's defining rhythm. Also: fun, which is a training variable.

9.1 Trail-sim fartlek

- **Structure:** on any run with even the tiniest undulation (bridges, dunes, park mounds), attack every rise at Z4–Z5 and float the flats. If your route is truly billiard-flat: 1 min hard / 2 min easy × 10–12, and *pretend*.
- **Why it works for trail:** effort variability on demand. The terrain (or the timer) decides, you respond — exactly the racing skill.

9.2 The Kenyan fartlek

- **Structure:** 1 min hard / 1 min easy × 15–20, continuous, effort by feel.
- **Why it works for trail:** a huge aerobic-power dose with no measuring, no track, no negotiating. The floats never fully recover you — hello, mid-race legs.

9.3 The minute ladder

- **Structure:** 1-2-3-4-3-2-1 minutes @ Z4–Z5, with half-duration easy jogs between.
- **Why it works for trail:** variable-length efforts, longer than they feel, shorter than you fear. A superb bad-weather session — no distances to measure.

9.4 The compound sessions — speed + climb in one hour

These combine flat speed with stair/incline work back-to-back, mimicking a hilly race's demands in ~50–60 minutes. They are the engine of Plan B's Tuesdays and excellent occasional sessions for anyone:

- **Combo 1 — "Ups and Overs":** 4 × 1km @ Z4 (2:00 rest), straight into 6 × 45s hard stair climbs or steep treadmill surges.
- **Combo 2 — "Short & Sharp":** 8 × 400m @ Z5 (75s rest), then 10 min continuous stair machine or 12% incline @ Z3.
- **Combo 3 — "The Sandwich":** 10 min Z4 tempo, 8 × 30s stair sprints, 10 min Z4 tempo.
- **Why they work for trail:** running fast on tired climbing legs and climbing on tired running legs is the whole sport, compressed.

9.5 The bad-weather matrix

No excuses left behind:

CONDITIONS	SESSION
Rain, wind, dark	Minute ladder or Kenyan fartlek (effort-based, no measuring)
Ice / storm	Treadmill: incline blocks (8.1), 30/30s, or broken tempo
Treadmill boredom	Compound sessions — the format changes every 10 minutes
Away for work	Hotel stairwell: Stairway A short version + strides in the corridor of shame

PART III

Strength & Technique

Elevation is a strength stimulus — if the terrain won't load your muscles, the gym must. Chapters 10–11: the two weekly strength sessions that build mountain legs indoors, and the trail skills you can drill on flat ground.

Chapter 10 — Strength & Durability: Building Mountain Legs Indoors

Elevation is, at its core, a strength stimulus. If the terrain won't load your muscles, the gym must. Two sessions per week, 30–40 minutes each, is enough to transform your race.

Session A — Climbing strength (heavy & slow)

EXERCISE	SETS × REPS	NOTES
Goblet or back squat	3 × 6–8	Heavy but clean; 3s lowering phase
Walking lunges	3 × 10/leg	Add dumbbells as they get easy
Step-ups (knee-height box)	3 × 8/leg	Drive through the heel; add weight over time
Single-leg Romanian deadlift	3 × 8/leg	Hamstrings + balance in one move
Calf raises (straight & bent knee)	3 × 15 each	Slow lowering — calves do huge climbing work

Session B — Descending resilience (eccentric & plyometric)

EXERCISE	SETS × REPS	NOTES
Slow eccentric squats (5s down)	3 × 6	The closest gym analogue to descending damage
Lateral lunges	3 × 8/leg	Side-to-side control for technical trail
Box step-DOWNS (slow)	3 × 8/leg	Control the lowering — this is descent training
Squat jumps or drop jumps	3 × 6	Land soft and quiet; builds braking capacity
Single-leg balance + reach	3 × 30s/leg	Eyes closed to progress; ankle insurance
Plank + side planks	3 × 40s	A stable trunk saves legs on rough ground

Timing: put strength on easy-run days or after quality sessions — never the day before a long run in peak weeks. Reduce to maintenance (one light session) in taper weeks.

How to progress across the plan

Strength follows the same periodization as the running:

- **Weeks 1–3 (learn):** light loads, immaculate form, full soreness expected after the first two Session Bs — that's the repeated bout effect being purchased (Chapter 2). Never schedule the first Session B within two days of a long run.
- **Weeks 4–10 (load):** add weight when a set feels easier than "hard but clean" — roughly every second week. The squats and step-ups should be genuinely heavy by week 8; bodyweight squats past week 4 are stretching, not strength.

- **Weeks 11–12 (hold):** same loads, no new exercises. Strength is banked; you're maintaining it.
- **Weeks 13–14 (taper):** one light session in week 13, nothing closer than 5 days to the race.

One dial at a time, as with the running: add load *or* reps *or* a harder variation, never all in the same session.

The no-equipment version

No gym, no excuses — the hotel-room translation:

INSTEAD OF	DO
Goblet squat	Backpack squat (load with books/water), same tempo
Weighted step-ups	Step-ups on a sturdy chair, rucksack on
Barbell calf raises	Single-leg calf raises off a step, slow lowering, ×15+
Drop jumps	Squat jumps, landing silently, on grass
Loaded lunges	Split squats, rear foot elevated, backpack on

Bodyweight done slow, single-leg, and to honest fatigue covers 80% of the need. The remaining 20% is why gyms exist.

Durability extras that cost almost nothing

- **Run on the softest, roughest ground you can find** — forest floor, grass, sand, dunes. Ankles and stabilisers adapt to chaos only by experiencing chaos.
- **Walk more.** Weighted-vest walks, long city walks, taking every staircase you meet. Time on feet is a stimulus.
- **Practice power-hiking form now:** hands on knees for very steep pitches, shorter quicker steps, forward lean from the ankles. On race day you should be able to hike at a pace that embarrasses runners who never practiced it.
- **Poles:** if the race allows them and you plan to use them, train with them on weighted hikes and at least three long runs. Poles used well save quads; poles used badly are expensive decoration. (Full technique: Chapter 11.)

Chapter 11 — Trail Technique School

Fitness gets you around the course; technique decides how much it costs. On flat roads, technique differences between runners are worth seconds. In the mountains they are worth *tens of minutes* — and more importantly, they decide whether your quads survive to the finish. Every skill below can be drilled on flat terrain or on the modest bumps, stairs, and rough ground you do have.

Uphill running

For gradients gentle enough to run (roughly under 10–12%, or "as long as running feels cheaper than hiking"):

- **Shorten your stride, keep your cadence.** Small, quick steps at your normal step rate. Overstriding uphill is the fastest way to cook calves and stall momentum.
- **Lean from the ankles, not the waist.** Imagine a straight line from ankle to hip to shoulder, tilted slightly into the hill. Folding at the waist collapses the lungs and loads the lower back.
- **Drive the arms.** Compact, purposeful arm swing — the arms set the rhythm the legs follow.
- **Look 5–10 metres up the trail,** not at your shoes. Your feet follow your eyes.
- **Flat drill:** run your strides and the final 200m of Kilometre Club repeats with exaggerated knee drive and arm swing. You are grooving the uphill pattern without the hill.

Power hiking — the most undertrained skill in trail running

Above ~12–15% gradient, hiking is genuinely faster per unit of energy than running, even for elites. A practiced hiker gains minutes on every long climb while breathing easier. Treat hiking as a discipline:

- **Two gears.** *Cruise:* upright, brisk, arms swinging — for moderate steepness. *Grind:* hands pressed on quads just above the knees, pushing down with each step — for the properly steep stuff. The arm push measurably unloads the legs.
- **Short steps win.** Big lunging steps up steep ground spike heart rate and torch glutes. Think staircase, not staircase-two-at-a-time — except when a big step avoids two awkward small ones.
- **Transition without drama.** Decide run-vs-hike by *effort*, not ego. The moment running pushes you out of Z2–Z3 on a climb, switch. Practice smooth transitions in your Stairway C sessions: 2 minutes climbing hard, 1 minute brisk hiking, repeat.
- **Benchmark it:** on a stair machine or long staircase, learn what your sustainable one-hour hiking effort feels like. That's your race-day climbing gear.

Descending — free speed, expensive mistakes

Descending is the skill flatlanders fear most and need most. The physics: every downhill step is a small jump landing, and braking hard multiplies the impact your quads must absorb eccentrically. Good descenders don't fight gravity; they negotiate with it.

- **Quick, short steps.** High cadence (think 180+) means each step carries less impact and each foot placement is a smaller commitment, easy to correct.
- **Lean down the hill, slightly.** Terrifying at first, but a slight forward lean keeps your feet landing under your hips. Leaning back turns every step into a heel-strike brake — the quad-shredder posture.
- **Arms wide and alive.** Elbows out like a tightrope walker. Ugly is fine; ugly and upright beats elegant and horizontal.
- **Eyes 3–5 metres ahead,** scanning for your next three foot placements. Trust peripheral vision for the step you're on. This is the core skill — descending is 80% eyes.
- **Brake early, in segments.** If a descent is long, control speed with a few deliberate slow-downs on the smoother parts rather than one long braking effort on the steep part.
- **Flat drills that transfer:** fast strides downhill on any gentle slope, grass or dune descents at play-speed, skipping and hopping drills, drop jumps and step-downs from Chapter 10, and short barefoot strides on grass for foot proprioception.

Technical ground: rocks, roots, mud, and sand

- **Rocks:** aim for the flat tops of stable-looking rocks; when in doubt, more smaller steps. Momentum is stability — the slower and more hesitant you go through a rock garden, the wobblier you become.
- **Roots:** treacherous when wet. Step *over*, never *on*, wet roots and rock slabs where possible; if you must, land with your foot pointing downhill-of-travel so a slip slides you forward, not sideways.
- **Mud:** shorten stride, lower your centre of gravity slightly, and run through the middle of shallow mud rather than the slick polished edges everyone else has smeared. Accept that some mud is a hiking section.
- **Sand and soft ground:** the best technical trainer flat regions offer. Dune or beach running builds ankle strength, forces quick light steps, and punishes overstriding — one session every week or two pays off directly.
- **The universal rule:** when terrain gets chaotic, *cadence up, stride down, eyes up*.

Poles: a masterclass in one page

If your race allows poles and the course has long, steep climbs, poles are worth carrying — **if trained**. Untrained poles cost more coordination and energy than they save.

- **Sizing:** with the tip on the ground, your elbow should bend at roughly 90°. For running-heavy races, some prefer 5cm shorter.
- **Uphill technique:** plant both poles together every two steps (the "double plant") on steep grinds, or alternate poles with opposite legs on moderate climbs. Push *down and back* through the straps — the strap, not your grip, transmits the force.
- **Downhill:** for most runners, poles go away on runnable descents (folded, stowed in the vest in under 20 seconds — practice this!). Use them only on very steep or slippery descents as outriggers.
- **Flat-terrain training:** use poles on every weighted hike (Stairway C) and at least three long runs, including practicing deploy/stow while moving. Your triceps and lats will announce that they, too, are climbing muscles.
- **Etiquette:** tips down and behind you near other runners; never fully extended sideways in a crowd; check the race rules — some races require carrying them the whole way if you start with them.

Cadence, posture, and the flat-runner's advantage

Flat training builds beautiful rhythm — protect it on the trail. The runners who suffer most in the mountains are those whose form dissolves when terrain breaks their rhythm. Practice *rhythm flexibility*: within one flat run, deliberately shift between cadences and stride lengths every few minutes. Your trail-sim fartlek (Chapter 9) is the perfect session for this. On race day, think of terrain changes as gear changes, not interruptions.

The drill menu

Ten minutes, twice a week, after easy runs. Rotate:

DRILL	DOSE	TRAINS
High knees, butt kicks, A-skips	2 × 20m each	Cadence, posture, stiffness
Sideways shuffle & carioca	2 × 20m each	Hip mobility, lateral control
Single-leg hops (grass)	2 × 10/leg	Ankle stiffness, landing control

DRILL	DOSE	TRAINS
Backwards jog	2 × 30m	Quad activation, balance
Barefoot strides (grass)	4 × 60m	Foot strength, proprioception
Curb taps / low step-downs	2 × 15	Descent pattern, quiet feet

PART IV

The Plans

Chapters 12–16: the periodization machinery, the rules for building your own weeks, and three complete plans — the full 14-week Plan A, the time-crunched Plan B, and the 10-week Plan C sharpener.

Chapter 12 — Periodization & Building Your Own Plan

The plans in Chapters 14–16 are complete — follow one and skip this chapter until your second race. But a plan you understand is a plan you can rescue when life interferes, and eventually you'll want to build your own. Here is the machinery.

The three-phase arc

Every plan in this book follows the same arc:

PHASE	LENGTH	JOB	FEEL
Base & Foundation	~5 weeks	Frequency, long-run habit, introduce all session types at gentle doses	"This is manageable"
Build	~5 weeks	Peak volume, longest runs, full-dose quality, fueling rehearsal	"This is a lot"
Peak & Taper	~4 weeks	Sharpen, absorb, shrink volume, keep intensity touches	"I'm ready and slightly paranoid"

Within each phase, load rises for 2–3 weeks then drops for a **recovery week** (~60–70% volume, quality reduced). The recovery weeks — 4, 8, and 13 in the 14-week plans — are where adaptation actually happens. **They are not optional**, and feeling good in one is not a reason to skip it; feeling good is what it's for.

The weekly skeleton

All plans share one skeleton. Build any week from it:

1. **One long run** (the cornerstone — if a week collapses, save this).
2. **One speed session** (Chapters 4–5: Kilometre Club, 400s, 30/30s...).
3. **One threshold or climbing session** (Chapter 6 tempo work, or Chapter 8 stairs).
4. **Easy running around them** (Z1–Z2, with strides twice a week).
5. **Two strength sessions** (Chapter 10), on easy days.
6. **At least one full rest day**. Two if you're over 45, injury-prone, or sleep-deprived — the plans survive it.

Never two quality days back-to-back. Never a hard day before the long run in peak weeks. Roughly 80% of weekly time stays Z1–Z2.

Progression rules

- **Volume:** raise weekly kilometres by no more than ~10% per week, and don't raise stair/incline volume and running volume sharply in the same week (Achilles insurance, Chapter 19).
- **Long run:** grows by 2km per week at most, with drop-back weeks; peaks at 24–26km two-to-three weeks out for a 30k race.
- **One dial at a time** in any session: more reps, or less rest, or faster — never all three.
- **If life happens and you miss days, never cram.** Skip, don't stack. Missed sessions are gone; the plan continues from today (rejoining rules: Chapter 20).

Scaling to other races

The skeleton scales. Adjust the long run and the climbing dose; the week structure stays identical.

RACE	LONG RUN PEAKS AT	CLIMBING DOSE	NOTES
15–20km trail, ~600m vert	18–20km	1 stair/incline session/ wk	Plan B minus 15%; more Chapter 5 speed
28–30km, ~1,000m vert	24–26km	1–2 sessions/wk	The plans as written
42km mountain marathon, ~1,800m+	30–32km + back-to-backs	2 sessions/wk, both strength days sacred	Plan A + 4 weeks: repeat weeks 7–10 pattern before the taper; fueling ceiling 80–90g/h becomes non- negotiable
50k+ / first ultra	by time: 3.5– 4h	2/wk + monthly real- trail day	Different book — but this skeleton still holds

The off-season / pre-plan base block

If the race is more than 14 weeks away, or your longest run is under 90 minutes, run this 4–6 week block first — it makes everything after it work better:

- 4–5 easy runs per week, one slightly longer (build to 90 min).
- Strides twice a week; one gentle fartlek (Chapter 9) weekly.
- Strength A and B weekly at learning loads.
- No tempo, no intervals, no heroics. Boring is the point.

Chapter 13 — Reading the Plans

The next three chapters are the complete plans. Conventions:

- Sessions are named from the Workout Library (Part II) — structures live there; Appendix A is the one-page cheat sheet.
- Volumes are **guides, not commandments**. Hit the key sessions (speed, climb, long); flex the rest.
- Rest days are written as rest. Walking, mobility, and gentle life count as rest. A "missed" rest day cannot be made up by resting harder later, and yes, people try.
- **Two weeks before race day, nothing new** — no new shoes, foods, or gear.
- Strength A/B (Chapter 10) run through every week of every plan: A early-week, B late-week or moved to Wednesday, maintenance-only in taper.

Chapter 14 — Plan A: The Full 14-Week Plan

Structure: 3 phases — Base & Speed Foundation (weeks 1–5), Build (weeks 6–10), Peak & Taper (weeks 11–14). Race day is at the end of week 14.

Weekly template:

DAY	SESSION
Monday	Rest or Strength A
Tuesday	Speed: Kilometre Club / Track Attack
Wednesday	Easy run + strides
Thursday	Tempo, Stairway Session, or trail-sim fartlek
Friday	Rest (+ optional Strength B if energy allows, otherwise move Strength B to Wednesday)
Saturday	Long run (the cornerstone)
Sunday	Easy recovery run or back-to-back medium run

Run/walk racers: rehearse your intervals in every long run from week 6 onward — e.g. 4:00 run / 0:45 walk on flats, switching to effort-based hiking on anything steep. The plan is fully compatible; simply apply your intervals to the easy and long-run portions.

Phase 1 — Base & Speed Foundation (Weeks 1–5)

Goal: consistent frequency, introduce all session types, build the long run to ~20km.

WK	TUE (SPEED)	THU (STRENGTH-ENDURANCE)	SAT (LONG)	SUN	~VOLUME
1	5×1km @ Z4, 2:30 rest	30 min Z3 tempo	14km Z2	6km Z1	38–42km
2	10×400m @ Z5, 90s rest	Stairway A: 10×45s	16km Z2	6km Z1	40–45km
3	6×1km @ Z4, 2:15 rest	35 min Z3 tempo	18km Negative Split (Z1→Z3)	7km Z1	44–48km
4	Recovery week: 4×1km @ Z4	25 min Z3	12km Z2	Rest	30–34km
5	12×400m @ Z5, 75s rest	Stairway C: 35 min weighted hike	20km Z2, last 3km Z3	7km Z1	46–50km

Phase 2 — Build (Weeks 6–10)

Goal: peak volume, long runs to 26km, back-to-back weekends, full fueling rehearsal on every long run.

WK	TUE (SPEED)	THU (STRENGTH- ENDURANCE)	SAT (LONG)	SUN	~VOLUME
6	6×1km @ Z4–Z5, 2:00 rest	Stairway B: 30 min continuous	22km Negative Split	8km Z1	50– 54km
7	14×400m alt. Z5/ Z6, 75s	40 min Z3–Z4 tempo	20km with 3×15 min @ Z3	10km Z2 (back-to-back)	54– 58km
8	Recovery week: 5×1km @ Z4	Stairway C: 30 min	14km Z2	Rest	36– 40km
9	7×1km @ Z5, 2:00 rest	Stairway A: 14×45s + strides	24km Negative Split	10km Z1–Z2	56– 60km
10	16×400m alt. Z5/ Z6, 60s	40 min Z4 tempo (broken: 3×12 min)	26km, last 5km @ Z3 – dress rehearsal: race shoes, vest, full fueling	8km Z1	58– 62km

Phase 3 — Peak & Taper (Weeks 11–14)

Goal: sharpen, absorb, arrive hungry.

WK	TUE (SPEED)	THU	SAT (LONG)	SUN	~VOLUME
11	8×1km @ Z5, 90s rest	Stairway C: 40 min weighted hike	22km with 4×10 min @ Z3–Z4	8km Z1	52–56km
12	6×1km @ Z5, 2:00 rest	30 min Z3 + 6 strides	18km Negative Split	6km Z1	44–48km
13	Taper: 4×1km @ Z4–Z5	20 min Z3 + 4 strides	12km Z2, last 2km Z3	Rest	30–34km
14	3×800m @ Z4, feeling smooth	20 min Z1 + 4 strides (Thu)	RACE DAY	—	Race + ~12km

Plan A rules

1. **The recovery weeks (4, 8, 13) are not optional.** Adaptation happens in the easy weeks, not the hard ones.
2. **Volume numbers are guides, not commandments.** Hit the key sessions (Tuesday, Thursday, Saturday); flex the rest.
3. **Every long run from week 6 onward is a fueling rehearsal.** Practice exactly what Chapter 17 prescribes.
4. **If life happens and you miss days, never cram.** Skip, don't stack. Consistency across 14 weeks beats any single heroic week.
5. **Two weeks before race day, nothing new** — no new shoes, foods, or gear.

Chapter 15 — Plan B: The Time-Crunched 14-Week Plan

Four runs a week, one strength session minimum, five to six hours total. This plan trades peak volume for ruthless efficiency: every run has a job, nothing is filler. It suits busy weeks, injury-prone bodies, and anyone whose goal is to finish a mountain 30k strong rather than race it to the second. Runners have finished far harder races on less.

What makes 4 days enough:

1. **Every quality session does double duty.** Speed and climbing-strength stimuli are combined into single "compound" sessions (Chapter 9.4).
2. **The long run is sacred.** It is the one non-negotiable; if a week collapses, save Saturday.
3. **Strength work carries more of the elevation load** — with fewer running kilometres, the Chapter 10 sessions become even more important, not less. Schedule both Session A and Session B weekly if possible; never fewer than one.
4. **Walking counts.** On rest days, take stairs, walk commutes, add an easy weighted walk. Cheap time-on-feet fills the volume gap without injury risk.

Weekly template:

DAY	SESSION
Monday	Rest (walk, mobility)
Tuesday	Compound quality: speed + climb work in one session
Wednesday	Strength A (no run)
Thursday	Easy run + strides, or Stairway session
Friday	Rest + Strength B (light)
Saturday	Long run
Sunday	Rest or 30–40 min recovery jog (optional)

The compound Tuesday sessions (full details in Chapter 9.4): Combo 1 "Ups and Overs", Combo 2 "Short & Sharp", Combo 3 "The Sandwich".

Phase 1 — Foundation (Weeks 1–5)

WK	TUE (COMPOUND)	THU	SAT (LONG)	~VOLUME
1	Combo 1 (3×1km version)	6km easy + 4 strides	12km Z2	26–30km
2	Combo 2 (6×400m version)	Stairway A: 8×45s	14km Z2	28–32km
3	Combo 1 full	7km easy + 6 strides	16km Negative Split	32–36km
4	Recovery: 3×1km @ Z4 only	5km easy	10km Z2	22–25km
5	Combo 3	Stairway C: 30 min weighted hike	18km Z2, last 2km Z3	34–38km

Phase 2 — Build (Weeks 6–10)

WK	TUE (COMPOUND)	THU	SAT (LONG)	~VOLUME
6	Combo 1 (5×1km version)	8km easy + 6 strides	20km Negative Split — full fueling from here on	38–42km
7	Combo 2 full	Stairway B: 25 min continuous	18km with 3×12 min @ Z3	38–42km
8	Recovery: 4×400m + 4×30s stairs	5km easy	12km Z2	26–30km
9	Combo 3	Stairway C: 35 min weighted hike	22km Negative Split	40–44km
10	Combo 1 (5×1km, faster)	8km easy + strides	24km, last 4km @ Z3 — dress rehearsal: race kit, shoes, full fueling	44–48km

Phase 3 — Peak & Taper (Weeks 11–14)

WK	TUE (COMPOUND)	THU	SAT (LONG)	~VOLUME
11	Combo 2 (10×400m version)	Stairway C: 35 min	20km with 3×10 min @ Z3–Z4	40–44km
12	Combo 1 (4×1km)	7km easy + strides	16km Negative Split	34–38km
13	Taper: 3×1km @ Z4	5km easy + 4 strides	10km Z2, last 2km Z3	24–26km
14	3×800m @ Z4, smooth	20 min Z1 + 4 strides	RACE DAY	Race + ~8km

Plan B rules

1. **The long run cannot be shortened two weeks in a row.** Everything else can flex.
2. **If you gain a fifth day some week, add an easy 6–8km jog — never a second quality day.**
3. **The two dress-rehearsal metrics** are the same as Plan A: by week 10 you should comfortably (a) take on 60g+ carbs per hour and (b) finish a long run faster than you started it. If either fails, fix it before adding anything else.
4. **On this volume, intensity discipline is everything.** With only four runs, the temptation is to make them all count. Resist — Thursday and the first two-thirds of Saturday stay genuinely easy.

Chapter 16 — Plan C: The 10-Week Sharpener

For the runner who is already fit — running 40km+ weeks comfortably, long run of 18km+ in the last month — but whose race arrived on short notice. Plan C skips the base phase (you've built it) and runs a compressed build + taper. **It is not a shortcut for the unfit:** if you don't meet the entry bar, run Plan B and adjust your race goals instead.

Entry bar: 4+ runs/week for the last 8 weeks; a comfortable 18km long run within the last 4 weeks; no current amber/red niggles (Chapter 19).

Weekly template: same as Plan A. Strength A + B from week 1 — at *learning* doses for two weeks if eccentric work is new to you, regardless of running fitness. Your tendons haven't read your training log.

Weeks 1–4 — Compressed build I

WK	TUE (SPEED)	THU	SAT (LONG)	SUN	~VOLUME
1	6×1km @ Z4, 2:00 rest	Stairway A: 12×45s	18km Negative Split	7km Z1	46–50km
2	12×400m @ Z5, 75s	35 min Z3–Z4 tempo	20km Z2, last 3km Z3	8km Z1	50–54km
3	7×1km @ Z4–Z5, 2:00	Stairway C: 35 min weighted hike	22km Negative Split	10km Z2 (back-to-back)	54–58km
4	Recovery: 4×1km @ Z4	25 min Z3	14km Z2	Rest	34–38km

Weeks 5–8 — Compressed build II

WK	TUE (SPEED)	THU	SAT (LONG)	SUN	~VOLUME
5	Wave tempo: 5×(3' Z4+/3' Z3)	Stairway B: 30 min	24km Negative Split	8km Z1	54–58km
6	14×400m alt. Z5/Z6, 60s	40 min broken tempo (3×12)	26km, last 5km @ Z3 — dress rehearsal: race kit, full fueling	8km Z1	56–60km
7	8×1km @ Z5, 90s	Stairway C: 40 min weighted hike	20km with 4×10 min @ Z3–Z4	8km Z1	50–54km
8	Recovery: 5×1km @ Z4	25 min Z3 + strides	16km Negative Split	Rest	38–42km

Weeks 9–10 — Taper & race

WK	TUE	THU	SAT	~VOLUME
9	4×1km @ Z4–Z5	20 min Z3 + 4 strides	12km Z2, last 2km Z3	30–34km
10	3×800m @ Z4, smooth	20 min Z1 + 4 strides	RACE DAY	Race + ~12km

Plan C rules

1. **The dress rehearsal moved to week 6** — that's your exam, four weeks out. Pass it (negative split, 60g+/h carbs, two-day recovery) or downgrade the race goal now, cheerfully.
2. **No cramming.** Ten weeks means missed sessions stay missed. The bar for adding "bonus" work is: never.

3. **Eccentric caution:** if Strength B and stairs are new, weeks 1–2 at half doses, and expect soreness. Your last hard eccentric exposure is week 7.
4. **Ten weeks is enough** — for the fit runner it's mostly specificity, sharpening, and rehearsal. Trust that the base you brought is the hard part, already done.

PART V

The Support Systems

Chapters 17–20: fueling with food you can actually cook, hydration and weather preparation, the injury traffic-light system, and recovery — the training that happens between sessions.

Chapter 17 — Fueling: Food You Can Actually Cook

Trail races over 2.5 hours are eating contests with running attached. The gut is trainable, and home-cooked fuel is cheaper, tastier, and gentler on the stomach than a pocket full of identical gels. Everything below is tested trail-runner fare you can make in a normal kitchen.

The numbers that matter

WHEN	TARGET
Night before	Big, familiar, carb-centred dinner; low fibre, moderate fat
Race morning (2.5–3h before)	100–150g carbs, low fibre, low fat
During the race	60–90g carbs per hour + 400–700ml fluid per hour + electrolytes
Within 60 min after	1.0–1.2g carbs/kg + 20–30g protein

Train the gut like a muscle: from week 6 onward, hit your in-race carb target on every long run. Nausea at 60g/h in July becomes comfort at 80g/h by race month — but only with practice.

Before — the night before

Keep it familiar and carb-heavy. Cook one of these:

- 1. Tomato rice (arroz de tomate style).** Soften an onion and 2 garlic cloves in olive oil, add 3 ripe grated tomatoes (or a can), a bay leaf, salt. Add 300g rice and 750ml stock; simmer until just soupy. Optional: white fish or a fried egg on top. Comfort food, huge glycogen load, almost no fibre.
- 2. Simple pasta al pomodoro with chicken.** Pasta, a light tomato-basil sauce, grilled chicken breast, parmesan. Skip cream-heavy sauces and big salads tonight.
- 3. Risotto with butternut squash.** Arborio rice, stock, roasted squash, a little parmesan. Gentle, energy-dense, easy to digest.

Avoid the night before: beans and lentils, giant raw salads, very spicy food, heavy fried food, and more than one glass of anything alcoholic.

Before — race morning (2.5–3 hours out)

- 1. Race-day oats:** 80g oats cooked in milk or water with a mashed banana, a big spoon of honey, pinch of salt. Add white bread + jam on the side if you need more.
- 2. Rice porridge:** leftover white rice simmered in milk with honey and cinnamon. Even gentler than oats (less fibre).
- 3. Pancake stack:** 3–4 plain pancakes with maple syrup or honey and a banana. Festive and effective.
- 4. The nervous-stomach option:** white toast with honey + a banana + sports drink. Minimal, safe, proven.

Then 20–30 minutes before the start: one gel or 2 dates, plus a few sips of water.

During — homemade race fuel

Mix and match to hit 60–90g carbs/hour. Cut everything into 2–3 bite pieces and wrap individually in foil or paper.

1. Sticky rice cakes (the cycling classic, perfect for trail). Cook 250g sushi rice. While warm, mix in 2 tbsp sugar or honey, a pinch of salt, and either (a) cinnamon + a little coconut milk, or (b) cream cheese + a drop of soy sauce for a savoury version. Press into a small tray, chill, cut into squares, wrap. ~20–25g carbs per square, easy to chew at Z2–Z3 effort.

2. Salted boiled baby potatoes. Boil, drain, toss generously with salt and a little olive oil. Sounds odd; tastes like salvation at kilometre 22 when sweet things have lost their appeal.

3. Stuffed dates. Medjool dates, pit removed, stuffed with a small dab of peanut butter and a flake of sea salt. ~18g carbs each, nature's gel.

4. Honey-maple homemade "gel". Mix 6 tbsp maple syrup or honey, 1 tbsp lemon juice, 2 pinches of salt, and 1–2 tbsp water to loosen. Carry in a small soft flask. ~25g carbs per 2 large gulps. Costs pennies.

5. Banana bread bites. A dense, low-fibre banana bread (white flour, extra banana, less butter than usual) cut into cubes. Real food that survives a race vest.

6. Drink your carbs. 500ml with 40g of maltodextrin+fructose mix (or simply sports drink powder) + electrolytes = fuel and hydration in one. On hot days this becomes the backbone of your strategy.

In-race rhythm: set a timer — eat something every 25–30 minutes from minute 30, sip fluids constantly, and take electrolytes (300–600mg sodium/hour, more in heat). Never wait to feel hungry; by then you are 30 minutes behind.

After — recovery meals worth finishing for

Within the first hour, then a proper meal within three.

- 1. The immediate fix (0–30 min):** chocolate milk, or a smoothie of banana + milk + oats + honey + a scoop of yogurt or protein powder.
- 2. Chicken & rice recovery bowl:** rice, shredded roast chicken, a fried egg, avocado, soy sauce, sesame. Carbs + ~35g protein + salt.
- 3. Salmon, sweet potato & greens:** roast both while you shower. Anti-inflammatory fats, potassium, protein.
- 4. Caldo verde, upgraded:** the classic potato-kale soup with double the usual chouriço and a side of good bread — salt, fluid, carbs, and comfort in one pot. Add white beans or shredded chicken for extra protein.
- 5. The big omelette:** 3–4 eggs, cheese, whatever vegetables survive the week, with toast and orange juice. The lazy champion of recovery meals.

Rehydration: weigh yourself before/after long runs occasionally. Drink ~1.5L per kg lost, with salt in the food or drink — plain water alone won't rehydrate you properly.

Caffeine: a strategy, not a habit

Caffeine reliably lowers perceived effort — roughly 2–4mg per kg of body weight, taken 45–60 minutes before you want it working.

- **Race morning:** your normal coffee with breakfast. Race day is not the day to skip it (hello, withdrawal headache) nor to double it (hello, portalo queue).

- **Mid-race:** save a caffeinated gel or cola-flavoured chew for the final third — the classic "second engine" at the point the race gets serious.
- **Practice it** on at least two long runs, and go easy on caffeine the two days before the race so race-day caffeine hits harder.

Eating through a training week

You don't need a spreadsheet diet; you need three habits:

1. **Fuel the work:** more carbs on and immediately around quality and long-run days (rice, pasta, potatoes, bread, fruit); slightly fewer on rest days, replaced with vegetables and protein. Never run key sessions fasted in this program — you are training the gut, not starving it.
2. **Protein every meal:** 1.6–2g/kg/day across 3–4 meals (eggs, dairy, fish, chicken, legumes) to repair what the eccentric work breaks down.
3. **Sunday batch habit:** once a week, cook a pot of rice, a tray of roast potatoes and vegetables, a protein, and one batch of race fuel (rice cakes or banana bread from the During section). Thirty minutes of cooking removes five bad weekday decisions and means every long run is fueled with tested food.

The race-week carb load, with numbers

Forget the giant pasta mountain the night before — that's just a bad night's sleep with extra steps. A proper load is 48 hours long and boringly gentle:

- **Two days out and the day before:** raise carbs to roughly 7–9g per kg of body weight per day (for a 70kg runner: ~500–630g — think generous rice/pasta/potato portions at every meal, plus juice, fruit, white bread, honey). Lower fibre and fat to make room.
- **Spread it across the day** — 3 meals and 2–3 snacks — rather than one heroic dinner.
- Expect to gain ~1–2kg of water weight with the glycogen. This is fuel, not fat; it will feel like magic at kilometre 24.

Chapter 18 — Hydration, Heat, Cold & Altitude

The mountain controls the weather; you control the preparation. This chapter is the environment manual.

Hydration & electrolytes: the longer version

Sweat is not just water. Depending on the person, a litre of sweat carries roughly 400–1,200mg of sodium, and mountain efforts of 3–5 hours can cost 2–5 litres. Replace water without salt and you dilute what's left — the road to heavy legs, sloshing stomach, and in extreme cases genuine danger.

- **Learn your sweat rate:** weigh yourself nude before and after a one-hour Z2 run (no drinking during). Each kg lost $\approx$ 1 litre/hour. Do it once in cool weather and once in heat.
- **In-race targets:** 400–700ml fluid per hour, scaled to heat and your sweat test; 300–600mg sodium per hour, up to 800–1,000mg for salty sweaters in heat (white crust on your cap is the tell).
- **Sources:** electrolyte tabs in your flasks, salted foods (the boiled potatoes earn their place again), or salt capsules on hot days.

- **The pee check:** pale straw in the days before the race = ready. Clear = overdoing it; add salt, not more water.

Heat: the flatlander's stealth advantage

Heat adaptation is one of the most powerful legal performance enhancers available, it takes about two weeks, and it requires no mountain. If your race region runs hot — or even might —:

- **In the final 4 weeks:** add overdressed easy runs (an extra layer on a mild day) or 15–25 minutes of post-run sauna or very hot bath, 2–3×/week. Adaptation arrives within 10–14 days: lower heart rate, earlier and saltier-then-diluter sweat, and a surprising boost to blood volume that helps even in cool weather.
- **Rules:** heat-adapt on *easy* days only, never around key sessions; hydrate aggressively afterwards; skip it entirely in race week except normal warm showers.
- **Racing in heat:** effort down a notch from the gun (heat taxes the same budget as pace), double the fluid plan, add sodium, pour water on your neck and wrists at aid stations, and treat shade like an aid station.

Cold and weather

Mountain weather can serve four seasons in four hours — the roadbook's mandatory jacket exists because people have died of exposure in summer races. Your flat-terrain homework:

- **Test the jacket** on a rainy training run: does it actually keep you functional, and can you get it on with cold hands, moving?
- **Layer rule for racing:** dress for the second hour, not the first ten minutes. Slightly cool at the start is correct.
- **Cold hands ruin races** — they can't open food wrappers. Light gloves weigh nothing; carry them if the summit forecast is under ~10°C.
- **Jacket on before you're cold** on exposed ridges (also Chapter 22's contingency list). Shivering is a late symptom, not an early warning.

Altitude, briefly

Most 30k mountain races top out below 2,000m, where altitude costs a few percent and needs no special preparation — pace by effort (which you were doing anyway) and it self-corrects. If your race lives higher: arriving either 2+ days early or the same morning both beat arriving the evening before; hydrate more; expect climbs to feel one zone harder than they are. True altitude racing (sustained 2,500m+) rewards a longer acclimatisation stay, which is a logistics decision, not a training one.

Chapter 19 — The Body Shop: Injury Prevention

The most common reason people miss their goal race isn't fitness — it's a six-week injury acquired in week nine. Almost every running injury announces itself politely before it shouts. This chapter is about hearing the announcement.

The traffic-light system

Use this for any new niggle, every day it persists:

LIGHT	DEFINITION	ACTION
Green	Mild ache that appears during warm-up and <i>disappears</i> as you run; gone next morning	Train normally, monitor
Amber	Pain that persists through a run without worsening, or lingers next morning; changes your stride slightly	Cut intensity and volume ~30–50%, swap a run for bike/ stair machine, start targeted rehab, reassess in 3–4 days
Red	Pain that worsens during a run, alters your gait, hurts walking, or wakes you at night	Stop running. Cross-train only if pain-free. See a physio if not clearly improving within a week

The rule that saves races: three amber days in a row equals a red day. And swelling, sharp bone-specific pain, or numbness is always red, immediately.

The flatlander-to-mountain injury watchlist

The training in this book adds two loads your body may not know: stairs/incline (calf and Achilles load) and eccentric strength work (quad soreness). Ramp both gradually — the plans already do, so don't "bonus" extra stair sessions in enthusiasm.

- **Achilles & calves:** the classic cost of new incline work. Prevention: the slow calf raises in Strength A, done religiously, plus never increasing stair volume and running volume sharply in the same week. Morning Achilles stiffness that eases with movement = amber.
- **ITB / outer knee:** often appears with volume jumps and fatigue. Prevention: the single-leg work and lateral lunges; respect recovery weeks. Pain on the outside of the knee that arrives at the same distance every run = amber, address early — ITB issues negotiate poorly once established.
- **Runner's knee (front of knee):** loves downhill and stair *descents*. Prevention: the step-downs and eccentric squats, plus building descent-simulation gradually.
- **Plantar fascia:** first-steps-of-the-morning heel pain. Prevention: calf strength, sensible shoe rotation, and not suddenly doubling barefoot or sand running.
- **Ankle sprains:** the trail classic. Prevention: the balance work in Strength B — thirty seconds per leg daily is cheap insurance — and eyes-up technique from Chapter 11. A rolled ankle in training: red until walking is painless.
- **DOMS vs. injury:** deep *symmetrical* muscle soreness 24–48h after eccentric work or long descents is DOMS — normal, trainable, and exactly the adaptation you want (it fades week by week; this is the "repeated bout effect" and it's your descent armour). Asymmetric, joint-specific, or sharp pain is not DOMS.

The 10-minute daily maintenance routine

Every day, ideally in the evening. It is deliberately short so you'll actually do it.

MINUTES	WORK
0–2	Calf raises off a step, slow lowering — 15 straight-knee + 15 bent-knee
2–4	Single-leg balance, 45s per side (progress: eyes closed, then on a cushion)
4–6	Hip openers: 10 deep lunges with rotation per side
6–8	Glute bridges ×15 + side-lying leg raises ×12 per side

Shoes, surfaces, and rotation

- **Rotate two pairs** if you can — different shoes distribute load differently, and the variety is quiet injury prevention.
- **Trail shoes:** buy them by week 8, break them in on at least four long runs. Grip appropriate to the terrain (dry rock and mud are different shoes). On dry, smooth trails, road shoes with decent tread survive; on anything wet, rocky, or steep and loose, grip failure is both slow and dangerous.
- **Surfaces:** softest, roughest available for easy runs; track or smooth path for speed; whatever the race serves for at least a few long runs.

Chapter 20 — Recovery, Monitoring & When Life Interrupts

Recovery is training

Adaptation happens between sessions, not during them. The recovery hierarchy, in order of actual importance:

1. **Sleep — the entire podium.** Aim for 7.5–9 hours in build weeks; treat the nights after long runs and quality days as sacred. One practical upgrade beats all gadgets: a consistent bedtime.
2. **Food and fluid timing** — Chapter 17's post-run window, honoured every time.
3. **Genuinely easy easy-days.** Z1 means Z1. Ego is the most common recovery deficiency.
4. **Movement snacks:** walking on rest days beats collapsing on the sofa; gentle movement clears soreness faster than immobility.
5. **The nice-to-haves:** massage or foam rolling (feels good, helps some), compression, cold water — use whatever you enjoy, but know they are the dessert, not the meal.

Monitoring: three numbers and a feeling

Every morning takes ten seconds: resting heart rate (or HRV if your watch does it), hours slept, and a 1–5 "legs and mood" score. You are not chasing precision — you are looking for *streaks*. Two or three days of elevated resting HR, poor sleep, and grumpy-heavy legs together = insert an extra easy or rest day *before* your body inserts one for you. A single bad number means nothing; ignore it.

Overreaching: the difference between tired and broken

Normal training tiredness lifts after a recovery week. Suspect real overreaching when, for two-plus weeks: performance drops despite effort feeling higher, sleep worsens *while* tired, resting HR stays elevated, and enthusiasm disappears (not just for running). The fix is never "train through it": take 5–7 genuinely easy days, eat properly (underfueling is the most common hidden cause, especially in runners quietly trying to lose weight mid-plan), and rebuild from where your body actually is.

When life or illness interrupts

- **Neck rule for illness:** symptoms above the neck (sniffles, mild sore throat) — easy running is fine if you feel like it. Anything below the neck (chest, fever, body aches, GI) — full rest until 48h symptom-free, then resume easy. Never do quality sessions with a fever, ever; the downside risk is wildly asymmetric.
- **Missed less than a week?** Rejoin the plan where it is now (not where you left it), trimming the first quality session.
- **Missed 1–2 weeks?** Repeat the last completed week at ~70% volume, then continue. Drop a race-week ambition level (A goal → B goal) and be at peace with it.
- **Missed 3+ weeks in the second half of the plan?** Rebuild long runs first, keep strides, park the hard intervals, and consider changing the race goal to "experience and finish". There will be other races; there is only one of you.

PART VI

The Race

Chapters 21–24: the mental game, race week hour by hour, the troubleshooting FAQ, and what happens after the finish line. Everything here was rehearsed in Parts I–V; race day contains zero experiments.

Chapter 21 — The Mental Game

A 30km mountain race is long enough to guarantee at least one genuinely bad patch — a stretch where the finish feels impossible, the legs feel borrowed from someone older, and quitting composes its opening arguments. This is not a character flaw. It is a scheduled event. Athletes who know it's coming treat it like weather: unpleasant, temporary, survivable.

Segmentation: never race 30 kilometres

The mind handles "the next 20 minutes" brilliantly and "three more hours" terribly. Before race day, cut the course into 4–6 chunks using the profile — climb 1, the ridge, the big descent, the rolling middle, the final climb, the run-in. For each chunk, write down:

- **A job** ("hike steadily, eat twice", "let the legs roll, don't brake")
- **A treat** (the stroopwafel you saved, the view, changing the playlist)
- **A checkpoint thought** ("after this descent I'm two-thirds done")

On race day you are never running a 30k. You are only ever doing the current chunk's job, collecting the treat, and starting the next one.

The bad-patch protocol

Post this in your brain now, retrieve it at kilometre 21:

1. **Eat and drink first.** Roughly 80% of mid-race despair is a fuel gauge misread as an existential crisis. Take a gel or two dates, big sips, then wait.
2. **Apply the 10-minute rule.** No decisions during a low. You may reassess your goals in 10 minutes; until then, just keep moving at whatever pace is available — hiking counts, hiking *is* racing.
3. **Shrink the world.** Next aid station. Next tree. Ten more steps of good posture. Repeat.
4. **Run a body scan and fix one thing:** drop the shoulders, unclench the jaw, shorten the stride. Physical fixes reroute the spiral.
5. **Deploy your mantra** (below), ideally aloud. Slightly mad, entirely effective.
6. **Remember the rule of patches:** every bad patch in a long race has ended. Yours will too, usually within 15–20 minutes, often the moment the sugar lands.

Mantras and self-talk

The research is boringly consistent: instructional and motivational self-talk measurably improves endurance performance, and second-person phrasing ("you") tends to beat first-person ("I"). Build two or three lines in training that feel true, short, and rhythmic:

- *"Smooth is fast."* — for descents and technical ground
- *"You've eaten hills for breakfast since July."* — for the base of the big climb
- *"Forward is a pace."* — for the bad patch
- *"Finish proud."* — for the last 5k

Cheesy? Absolutely. So is everything at kilometre 26, and yours will be the useful kind of cheese.

Train the head like the legs

- **Rehearse discomfort deliberately.** The final third of every Negative Split long run is your weekly appointment with race-day feelings. When it gets hard, practice the protocol above instead of just enduring — the skill is *responding*, not suffering.
- **Do one long run solo without music.** Boredom tolerance is an endurance discipline, and race-day quiet stretches will feel familiar instead of hostile.
- **Visualise usefully:** twice in race week, spend five minutes walking mentally through the race — including the hard parts going *manageably*, you executing the protocol, and the finish. Visualising only glory prepares you for nothing; visualising problems-plus-responses prepares you for everything.
- **Keep a confidence bank.** One line per key session in a notes file: what you did, how you coped. In week 14, when the taper gremlins whisper that you're not ready, read the ledger. Fourteen weeks of receipts out-argue any gremlin.

Goals that survive contact with a mountain

Set three before race week, write them down, tell someone:

- **A — the dream day:** everything clicks (a time or a placing).
- **B — the solid day:** the realistic target that fourteen weeks earned.
- **C — the proud finish:** finish healthy, having fueled and paced like an adult.

Mountains veto plans via weather, stomachs, and surprise mud. Runners with one goal have a 30km opportunity to fail; runners with three have a guarantee that something remains worth fighting for at kilometre 22. That difference is worth more than any single training session in this book.

Chapter 22 — Race Week & Race Day

Pacing a mountain 30k as a flatlander

1. **Throw pace out the window; race by effort.** Your GPS pace is meaningless on a climb. Aim for Z2 breathing everywhere, allowing Z3 briefly on climbs. If you're breathing like a Tuesday interval session in the first hour, you will pay with interest.
2. **Hike the steep stuff from kilometre one.** Even the front of the field hikes steep gradients — it's faster per heartbeat. You trained this on the stairs; use it proudly.
3. **Bank energy, not time.** The flatlander instinct is to hammer the runnable early sections. Resist. The race begins at the two-thirds mark; everything before is transport.
4. **Descend with quick, short steps,** slight forward lean, eyes 3–5 metres ahead, arms wide for balance. Braking with long strides is what shreds quads. Your eccentric strength work bought you resilience — spend it wisely and late.
5. **Run/walk racers:** hold your rehearsed intervals on flat and rolling ground, switch to terrain-based decisions (hike steep ups, run everything runnable) when the course gets serious. Walk breaks are a strategy, not a failure — used from the start they're worth many minutes at the end.
6. **Smile at kilometre 20.** Genuinely. It lowers perceived effort. Science says so, and it annoys the competition.

Kit checklist

- Trail shoes **broken in over at least 4 long runs** — grip appropriate to the terrain (dry rock vs. mud are different shoes)
- Race vest with soft flasks — worn on every long run from week 6
- Mandatory kit per the race rules (jacket, blanket, phone, cup) — check the roadbook twice
- Anti-chafe balm everywhere you have ever chafed, plus one place you haven't yet
- Cap/buff + sunscreen; mountain weather can serve four seasons in four hours — pack the jacket even if the valley is sunny
- Light gloves if the summit forecast is under $\sim 10^{\circ}\text{C}$ (cold hands can't open food)
- Poles if permitted and trained with (see Chapter 11)

General tips that earn their place

1. **Sleep is the best legal performance enhancer.** Prioritise it in build weeks over any extra session.
2. **Study the course profile until you can draw it from memory.** Break the race into 4–5 chunks with a plan and a treat (food, music, view) for each.
3. **Heat-train if your race region runs hot:** in the final 4 weeks, add overdressed easy runs or post-run sauna 2–3×/week (full protocol: Chapter 18).
4. **Do at least one long run on the most technical terrain you can reach,** even if it needs a train ride. One real trail day teaches feet more than five park loops.
5. **Practice everything:** breakfast, kit, fuel, pacing, even the pre-race coffee timing. Race day should contain zero experiments and zero surprises.
6. **Beware the taper crazies.** Weeks 13–14 you'll feel sluggish, phantom injuries will appear, and you'll be convinced fitness is evaporating. It isn't. It's arriving.
7. **Set three goals:** A (dream day), B (solid day), C (finish proud). Mountains have opinions; flexible runners have good races.
8. **Respect the descent cutoffs of your own body.** If quads start cramping, shorten stride, add salt, and hike a few minutes early rather than blowing up completely.

Race week, day by day

DAY	TRAINING	EVERYTHING ELSE
Mon	Rest or 20 min jog	Check the roadbook: mandatory kit, cutoffs, aid station locations and what they serve. Trim toenails (yes, today — not Friday).
Tue	Short sharpener (3–4 reps, smooth)	Charge/test the watch, download the course GPX, study the profile and mark your segments (Chapter 21).
Wed	20–30 min easy + strides	Begin sleep banking — the night two days before matters more than the night before. Shop for carb-load food and race fuel.
Thu	Rest or 20 min jog	Carb load begins (Chapter 17). Cook and wrap your race food. Lay out full kit and photograph it.
Fri	15–20 min leg-loosener + 4 strides, or rest	Carb load continues. Travel/bib pickup. Pack the vest completely. Early, salty dinner from the familiar list. Set two alarms.
Sat	Race	Execute. Smile at kilometre 20.

The race-morning timeline

Assuming a morning start — shift as needed:

- **T-3h:** wake, coffee, race breakfast from Chapter 17. Begin sipping 500ml electrolyte drink.
- **T-2h:** dress, lube every chafe point, tape anything that needs taping. Final kit check against the photo.
- **T-1h15:** leave for the start with margin for the toilet queue, which is a mandatory aid station of its own.
- **T-40min:** drop bag if applicable. 8–10 min easy jog + 4 gentle strides *only if* racing hard from the gun; for most, brisk walking and a few leg swings suffice — the first climb is your warm-up.
- **T-20min:** one gel or 2 dates, last sips, stow flasks.
- **T-5min:** seed yourself honestly in the pen — mid-pack flatlanders who start behind their fitness pass people all day, which is the most enjoyable race strategy ever devised.

Aid station protocol

Racers lose whole minutes dithering at tables. Decide your routine now:

1. Arrive knowing what you need (you memorised the roadbook): typically **refill flasks, grab 2 things for the pocket, eat 1 thing walking out**.
2. Never stand and graze. Walk the length of the station while doing your tasks; it doubles as a scheduled walk break.
3. Thank the volunteers — it's free serotonin and they are the best people on the mountain.
4. Do a 10-second systems check leaving each station: food since last hour? Salt? Any hotspot in the shoes that needs 60 seconds of attention *now* rather than a blister at 25k?

Contingency lines

- **Weather turns:** jacket on *before* you're cold on exposed ridges; effort up slightly in cold rain, pace down in heat with doubled fluids and salt.
- **Stomach revolts:** drop to sips of plain water and slow for 15 minutes, then restart fueling with your blandest option (potatoes over gels). A settled stomach at slow pace beats a rebellion at goal pace.
- **Cramp twinges:** immediate salt + shortened stride + 2 minutes of hiking; stretch the offender gently only if it fully seizes.
- **Blister hotspot:** fix at the next stop, always. Sixty seconds of tape saves sixty minutes of limping.
- **Missed a fueling window and feel hollow:** two gels 10 minutes apart, hike a climb while they land, resume. You are 20 minutes from feeling reborn — every experienced trail runner has taken this exact ride.

Chapter 23 — Troubleshooting & FAQ

"My 1km repeats keep getting slower across the session. Broken?" Usually one of three things: starting the first rep too fast (run the first like it's the fifth), insufficient warm-up, or accumulated fatigue — check whether your easy days are actually easy. Slowing 2–3% across a session is normal; falling off a cliff is a pacing or fatigue signal, not a fitness one.

"I can't hit Zone 2 — my easy pace feels embarrassingly slow." Correct. That embarrassment is the sensation of doing it right. Flat-terrain runners with speed backgrounds chronically run easy days too fast. Slow down until

conversation is genuinely comfortable; within four to six weeks the same effort gets faster on its own. Nobody at the mountain race will ask about your Wednesday pace.

"My long runs are flat and mind-numbing. The race won't be." Three fixes: run them by *time* not distance so the watch stops taunting you; build in structure (the Negative Split and Z3 blocks exist partly as entertainment); and once or twice in the plan, travel — one train ride to real trails or dunes is worth five identical park loops for feet, head, and morale.

"Stairs bore me / I have no stairs / my gym has no stair machine." Ranked substitutes: treadmill at 10–15% incline (excellent), any bridge/overpass/parking-garage ramp repeated shamelessly (good), weighted-vest brisk walking (good for hiking economy), cycling hard in a big gear at low cadence (decent strength stimulus, zero impact). Combine two.

"I'm mid-plan and exhausted all the time." Check in order: sleep hours, easy-day discipline, eating enough carbs around sessions, and whether you skipped the recovery weeks. If all four are honest and heaviness persists past a full recovery week, downgrade to Plan B's volume for two weeks — arriving fresh beats arriving fit. (Full overreaching checklist: Chapter 20.)

"Race is hillier/less hilly than I trained for — panic?" No. The skills are transferable in both directions. Hillier: lean harder on hiking discipline and start even more conservatively. Flatter: your speed work becomes the weapon and the race just got friendlier to you specifically.

"Can I race in my road shoes?" On dry, smooth trails, road shoes with decent tread survive. On anything with mud, wet rock, or steep loose descents, trail shoes are not a luxury — grip failure is both slow and dangerous. If buying one pair: a moderate-lug all-rounder, bought by week 8, worn on at least four long runs.

"I only have a 12-storey stairwell / one small bridge. Enough?" Yes — repetition replaces length. Twelve storeys climbed twenty times is a mountain; the only cost is boredom, and Chapter 8's session variety exists to manage it. What repetition can't fake is long descents, which is why Strength B and the descent-simulation menu (8.5) matter doubly for you.

"Should I train fasted to burn more fat?" Not in this program. The race is carb-limited, the gut needs training, and fasted quality work mostly teaches you to run slower. Fat adaptation happens plenty at Z1–Z2 with normal eating.

"How much does a watch matter?" Less than the industry hopes. Effort zones, a timer for fueling alerts, and the course GPX on race day cover 95% of the value. The most important instrument in this book is the talk test, which ships with your mouth.

"How do I know the plan worked?" The dress rehearsal is the exam: 24–26km with the final kilometres *faster* than the start, 60g+ carbs per hour with a calm stomach, and legs that recover within two days. Pass those three and the mountain holds no surprises your training hasn't previewed.

Chapter 24 — After the Finish Line

The first 48 hours

Eat the celebration meal without arithmetic — you earned it and recovery requires it. Keep moving gently (the evening shuffle-walk after a mountain race is comically stiff and genuinely helpful), sleep long, and expect DOMS

to peak around hour 36, especially descending stairs backwards-worthy quads. Skip anti-inflammatories in the first day if you can; the inflammation is the repair crew.

The reverse taper (four weeks)

WEEK	RUNNING	NOTES
+1	0–3 very easy short jogs, only if the body volunteers	Walks, swims, naps. No watch targets.
+2	3–4 easy runs, ~50% of normal volume	Add strides at the end of one run if legs feel springy
+3	~70% volume, one light fartlek	Resume strength work at half loads
+4	Normal easy volume	You may now have opinions about the next race

The engine recovers before the chassis: your cardiovascular system will feel ready days before tendons and muscles actually are. This mismatch is the classic post-race injury trap — let the slowest system set the calendar.

The post-race blues are real

A strange flatness often arrives 3–10 days after a goal race, even a triumphant one. Fourteen weeks of structure and purpose ended in a single morning; some deflation is the normal exhaust of a big goal. Plan for it: book something pleasant in week +1, write your race report while details are fresh (future-you will treasure it), and don't sign up for anything in the first 72 hours of either euphoria or despair.

The debrief

A week after, answer five questions in writing: What went right that I planned? What went right that was luck? What failed, and was it preparation, execution, or mountain? What would I change in the next block? And the big one — did I enjoy this enough to want more of it? Your next training block should be built on those answers, not on the finish time alone.

Where to next

You now own a complete method: zones you can feel, a workout library, mountain legs built indoors, and a rehearsed race process. It scales (Chapter 12): the same skeleton, with a longer long run and a heavier climbing dose, reaches a mountain marathon; with a lighter one, it makes you dangerous at a fast 20k. Keep the strength work year-round at one session a week — it is the cheapest fitness you own — and keep one real-trail day a month if you can, purely because it's the point.

You also own something no flat horizon can take away: proof that the mountain was never the obstacle. It was the excuse to become this version of you.

Closing — One Flat Kilometre at a Time

Flat training is not a handicap — it is a different syllabus. Runners from the lowlands arrive at mountain races with speed, discipline, and rehearsed fueling that mountain locals often lack. Do the stairs. Lift the weights. Eat on

every long run. Practice the walking as seriously as the running. Start slower than pride wants, descend smarter than fear suggests, and keep three goals in your pocket.

And when the trail finally tilts upward on race day, you'll discover the climb was never the hard part — showing up prepared was, and you already did that, one flat kilometre at a time.

Boa sorte — see you on the mountain.

APPENDICES

The Fridge Door

The one-page versions: the complete workout quick reference, zone anchor pace tables, season planning, and the glossary.

Appendix A — Workout Library Quick Reference

The fridge-door version of Part II. Structures are abbreviated; full details at the chapter given.

Speed & VO2max (Chapters 4–5)

SESSION	STRUCTURE	PURPOSE
Strides (4.1)	4–8 × 20–30s relaxed fast, full recovery	Free speed, form — twice weekly
Flying 30s (4.2)	6–8 × 30m near-max from rolling start	Mechanics, recruitment
Track Attack (4.3)	8–16 × 400m @ Z5–Z6, 60–90s	Leg speed, economy, gear changes
200m turnover (4.4)	12–16 × 200m @ Z6, 200m jog	Cadence; taper-friendly short version
Stair sprints (4.5)	8–12 × 20–30s max stairs, full rest	Climbing power (hill sprints, substituted)
The Kilometre Club (5.1)	4–8 × 1km @ Z4–Z5, 90s–2:30 jog	Sustained hard efforts; climb-length surges
800s (5.2)	6–10 × 800m @ Z5, 400m jog	VO2 volume with form retention
Billat 30/30s (5.3)	12–20 min of 30s Z5 / 30s jog	VO2max anywhere, no track
The Ladder (5.4)	400-800-1200-800-400 @ Z5	Variable-length efforts, commitment
3-min repeats (5.5)	5–7 × 3:00 @ Z5 by feel, 2:00 jog	Effort-based pacing clock
Cut-downs (5.6)	6 × 1km, Z4 → Z5+, each faster	Acceleration on tired legs

Threshold (Chapter 6)

SESSION	STRUCTURE	PURPOSE
Classic tempo (6.1)	20–40 min @ Z4 continuous	The long-climb simulator
Cruise intervals (6.2)	3–5 × 8–12 min @ Z4, 2' jog	More quality, better form
Wave tempo (6.3)	4–6 × (3' Z4+ / 3' Z3), continuous	Absorbing surges — most race-specific
Progressive tempo (6.4)	30–40 min, Z3 → Z4+ ramp	Negative-split discipline
Broken tempo (6.5)	2 × 15 min @ Z4, 3' jog	Time-crunched threshold
Hour of power (6.6)	3 × 15 min @ Z4 — once, wk 9–11	Peak-phase graduation exam

Long runs (Chapter 7)

SESSION	STRUCTURE	PURPOSE
The Negative Split (7.1)	Long run in thirds: Z1→Z2→Z3	Strength on tired legs; pacing discipline

SESSION	STRUCTURE	PURPOSE
The Sandwich (7.2)	Long run with 3–4 × 10–15 min @ Z3–Z4	Mid-race climbing rehearsal
The Surge run (7.3)	Easy long + 8–12 × 1 min @ Z4 random	Terrain-decides gear changes
Back-to-back (7.4)	Sat long + Sun medium, both easy	Hours-4-and-5 durability
Fast-finish (7.5)	Easy long, final 3–5km @ Z3–Z4	The last-5k gearshift
Time-on-feet (7.6)	3h+ by time, rough ground, hike slopes	Feet, ankles, morale, real-trail day

Elevation substitutes & mixed (Chapters 8–9)

SESSION	STRUCTURE	PURPOSE
Stairway A — Power (8.2)	10–15 × 30–60s hard stair climbs	Climbing power
Stairway B — Endurance (8.3)	20–40 min continuous stairs/machine @ Z3	Sustained climbing
Stairway C — Weighted hike (8.4)	30–45 min, 4–6kg pack, stairs/steep incline	Hiking economy — the race-day gear
Downhill treadmill (8.5)	10–20 min easy @ –2 to –5%, fortnightly	Descent armour (repeated bout effect)
Sand & dunes (8.7)	Easy runs on soft rough ground	Ankles, quick feet, honorary hills
Trail-sim fartlek (9.1)	Attack every rise Z4–Z5, float flats; or 1' hard / 2' easy × 10–12	Effort variability
Kenyan fartlek (9.2)	15–20 × 1' hard / 1' easy, continuous	Big aerobic power, no measuring
Minute ladder (9.3)	1-2-3-4-3-2-1 min @ Z4–Z5	Bad-weather quality
Ups and Overs (9.4)	4×1km @ Z4 + 6×45s stairs	Compound speed+climb (Plan B Tue)
Short & Sharp (9.4)	8×400m @ Z5 + 10 min incline @ Z3	Compound speed+climb (Plan B Tue)
The Sandwich combo (9.4)	10' Z4 + 8×30s stairs + 10' Z4	Threshold under fatigue (Plan B Tue)

Strength (Chapter 10)

SESSION	CONTENT	PURPOSE
Strength A	Squats, lunges, step-ups, SL RDL, calf raises	Climbing strength — weekly
Strength B	Eccentric squats, step-downs, jumps, balance, core	Descending resilience — weekly
Daily 10-min routine (Ch. 19)	Calves, balance, hips, glutes, stretch	Injury insurance — daily

Appendix B — Zone Anchor Tables

Anchor zones off a recent 5k time trial (Chapter 3). Paces are for flat ground; on any gradient, revert to effort and the talk test.

5K TIME	Z2 EASY (/KM)	Z3 STEADY	Z4 THRESHOLD	Z5 (≈5K PACE)
18:00	4:50–5:35	4:20–4:45	3:50–4:10	3:36
20:00	5:15–6:00	4:45–5:10	4:15–4:35	4:00
22:00	5:40–6:25	5:10–5:35	4:40–5:00	4:24
24:00	6:05–6:50	5:35–6:00	5:05–5:25	4:48
26:00	6:30–7:15	6:00–6:25	5:30–5:50	5:12
28:00	6:55–7:40	6:25–6:50	5:55–6:15	5:36
30:00	7:20–8:05	6:50–7:15	6:20–6:40	6:00

Z1 is slower than Z2 and cannot be too slow. Z6 is faster than 5k pace and self-limits. If your 5k falls between rows, split the difference; if your easy pace from the table feels hard, believe the feeling and slow down — the table serves the talk test, not the reverse.

Race-effort translation for a mountain 30k: flat sections at Z2 feel, climbs capped at Z3 breathing (hike when running exceeds it), descents at whatever effort keeps you relaxed and quick — usually Z2 breathing with Z4 concentration.

Appendix C — Season Planning

One goal mountain race is a project; a season is a rhythm. A sane year for the flat-terrain trail runner:

BLOCK	LENGTH	CONTENT
Off-season / base	4–8 wks	Easy volume, strides, weekly fartlek, strength at learning loads (Ch. 12)
Plan (A/B/C)	10–14 wks	This book, as written
Race + reverse taper	4 wks	Chapter 24
Bridge	2–6 wks	Easy running, one quality session/wk, keep strength — or roll into the next base

- **Two goal mountain races a year is plenty.** Three is possible with short plans (Plan C) and honest recovery; four is how people meet physios.
- **Tune-up races:** a flat 10k or half in weeks 5–8 of a plan slots in as a quality session and re-anchors your zones — take the following week's Tuesday easy.

- **Keep year-round, even off-plan:** strides twice a week, one strength session, one longer run. Those three preserve 80% of the machine at 30% of the cost.
-

Appendix D — Glossary

- **Back-to-back:** medium-long run the day after the long run, training endurance on pre-fatigued legs.
- **Cadence:** steps per minute; higher cadence with shorter steps is the safe pattern on descents and technical ground.
- **Cruise intervals:** threshold work broken into reps with short floats (Chapter 6.2).
- **Cutoff:** a time limit at a checkpoint after which runners may not continue. Know yours from the roadbook.
- **DOMS:** delayed-onset muscle soreness, peaking 24–48h after unfamiliar (especially eccentric) load; fades with repetition.
- **Eccentric:** muscle working while lengthening — quads during descents and slow squat lowering. The main source of mountain-race leg damage and the main target of Strength B.
- **Fartlek:** unstructured speed play — surges and floats by feel.
- **Lactate threshold:** the hardest effort you can sustain roughly an hour; the ceiling this book keeps raising (Chapter 6).
- **Negative split:** finishing faster than you started; the signature of a well-paced effort.
- **Over-unders:** threshold reps alternating slightly above and below threshold (Chapter 6.3).
- **Power hiking:** deliberate fast walking on steep gradients, often hands-on-quads; faster per heartbeat than running above ~12–15%.
- **Repeated bout effect:** the protection against eccentric muscle damage granted by a prior exposure; lasts weeks (Chapter 2). Your descent armour.
- **Roadbook:** the race's official document of course, aid stations, mandatory kit, and cutoffs. Read it twice.
- **RPE:** rating of perceived effort; the zones in this book are RPE with better marketing.
- **Strides:** short relaxed accelerations for speed and form, not a workout in themselves.
- **Taper:** the deliberate volume reduction before racing, during which fitness arrives disguised as anxiety.
- **Technical (terrain):** ground that demands foot-placement attention — rocks, roots, ruts, scree.
- **Vert:** total elevation gain, in metres. The reason this book exists.
- **VO2max:** maximal rate of oxygen use; the roof over every other intensity (Chapter 5).